Chao Cao

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SUMMARY

I am a Computer Science and Engineering Ph.D. candidate at the University of Texas at Arlington. My research focuses on scalable PyTorch-based machine learning pipelines for large-scale MRI processing, medical imaging analysis, neurodegenerative disease prediction, and multimodal healthcare AI. My work has appeared in AAAI, MICCAI, and ISBI. I am currently exploring medical agentic AI systems involving LLM fine-tuning, LLM post-training, GraphRAG retrieval, and LangGraph-based workflow orchestration.

EDUCATION

University of Texas at Arlington | ***Ph.D.*** *in Computer Science and Engineering* **Jan. 2023 – Dec. 2026(Expected)**

University of Maryland at College Park | ***Master*** *in Data Science and Analytics* **Jan. 2021 – Dec. 2022**

Nanjing University of Posts and Telecommunications | ***Bachelor*** *in Computer Science and Technology* **Sep. 2017 – Jul. 2021**

WORK EXPERIENCE

Software Development Engineering Intern | GyriQAI, Inc. **May. 2025 – Present**

Gemini API, GPT-4o, LLM Orchestration, Python, asyncio, Playwright, Docker, SQLite *Athens, GA*

- Designed a **multi-model LLM orchestration system** integrating Gemini and GPT-4o across **8+ pipelines**, with few-shot prompting, tier-based routing, quality scoring, confidence-guided filtering, and YAML-driven prompt templates for scalable content generation.
- Built agentic automation and data curation pipelines, including a **4-agent** social media framework and synthetic data workflows processing **94GB** across **50K+ schools**, producing **1M+** structured SFT samples.

Research Assiatant | University of North Carolina at Chapel Hill, BRAIN Lab **May. 2024 – Aug. 2024**

AI/ML, Medical Imaging, Cortical Surface Modeling *Chapel Hill, NC*

- Led the development of **Deep-GyralNet**, the first Spherical U-Net-based PyTorch framework for learning-based GyralNet extraction and 3-hinge gyrus identification, resulting in a first-author ISBI paper[1].
- Combined Focal loss, Connectivity-Aware Path Enforcement loss, and confidence-guided geodesic refinement, achieving 71.64 Dice, 98.20 recall, 96.40 F1-score, and reducing runtime from 237s to 14s per subject.

RESEARCH & PROJECT EXPERIENCE

Multi-Agent Clinical Decision Support System **Feb. 2026 – May. 2026**

LangGraph, FastAPI, GraphRAG, PostgreSQL, Docker *Arlington, TX*

- Developed a LangGraph-based 5-agent clinical AI system with StateGraph orchestration, conditional routing, shared memory/state, and traceable execution for end-to-end clinical decision support.
- Integrated GraphRAG retrieval, HIPAA-oriented PHI de-identification, ICD-10 coding, and drug-drug interaction detection into FastAPI services with Dockerized deployment, achieving **94% diagnostic accuracy**.

Multimodal LLM Post-Training for K12 Math Reasoning **Jan. 2026 – May. 2026**

Qwen2.5-VL-32B, PyTorch, Megatron-LM, verl, vLLM, CPT/SFT/RL *Arlington, TX*

- Designed a scalable **CPT → SFT → RL** post-training pipeline for **Qwen2.5-VL-32B**, using Megatron-LM to train on **20M** image-text pairs and 100B text tokens, followed by full-parameter SFT on **770K** reasoning samples.
- Built a GSPO-based RL stage with **verl**, **vLLM** inference acceleration, curated **80K** high-quality samples, and rule-based rewards with **98.5%** verification accuracy, improving image-based math accuracy by up to **+20%** and text-only accuracy by **+12%**.

Radiology NLI Benchmarking with Large Language Models [4] **Nov. 2023 – Jun. 2024**

LLM Evaluation, Medical NLP, Benchmarking *Arlington, TX*

- Built **RadQNLI**, a **9,838**-pair radiology NLI benchmark transformed from RadQA, to evaluate LLM reasoning over clinical reports and question-specific answer evidence.
- Designed prompt-based evaluation protocols for ChatGPT/GPT-4 and compared zero-shot, few-shot, CoT, and semantic-alignment prompting against open-source LLMs and fine-tuned BERT-family baselines.

PUBLICATIONS

Research Impact: 900+ citations, h-index 10, i10-index 11 — Google Scholar

1. **Cao, C.***, Cheng, J.*, Chen, M., Zhang, J., Liu, T., Zhang, L., et al. (2026). Deep-GyralNet: Enabling Machine Learning in Gyral Folding Pattern Extraction on Cortical Surface. *IEEE International Symposium on Biomedical Imaging (**ISBI 2026 Oral**)*
2. **Cao, C.***, Chen, T.*, Zhao, N.*, Chen, M., Qu, M., Zhang, Z., Shi, X., Li, X., Liu, T.,et al. (2026). Gyral-Sulcal-Net: An Integrated Network Representation of Brain Folding Patterns. *IEEE International Symposium on Biomedical Imaging (**ISBI 2026 Oral**).*
3. **Cao, C.**, Yu, X., Zhang, L., Chen, T., Liu, T., Zhu, D, et al. (2024). Enhancing group-wise consistency in 3-hinge gyrus matching via anatomical embedding and structural connectivity optimization. *IEEE International Symposium on Biomedical Imaging (ISBI 2024).*
4. Wu, Z.*, Zhang, L.*, **Cao, C.***, Yu, X., Dai, H., Ma, C., Liu, Z., Zhao, L., Li, G., Liu, W., et al. (2024). Exploring the trade-offs: Unified large language models vs local fine-tuned models for highly-specific radiology NLI task. *IEEE Transactions on Big Data.*
5. Cheng, H., Yu, X., Wu, S., Fang, L., **Cao, C.**, Zhang, J., Liu, T., Zhu, D., et al. (2026). DCMM-Transformer: Degree-Corrected Mixed-Membership Attention for Medical Imaging. *Proceedings of the AAAI Conference on Artificial Intelligence (**AAAI 2026**).*